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“INDIA IS STRATEGIC FOCUS FOR EMBEDDED BUSINESS”

Sanat Rao, marketing director of Intel's embedded and communication group, talks to EFY's Shweta Dhadiwal on Intel's embedded business development in India. In this interaction, he explains the Indian embedded market, the kind of support that Intel provides to the embedded community and some design tips for the developers

What are your views on the Indian embedded market and development community here?

We see India as one of our focus areas for the embedded business. The local embedded market is rapidly growing in terms of consumption as well as design. We see a lot of unique India-specific products like portable embedded devices for the financial inclusion market. A large part of embedded development in the world is done in India. Lots of our international customers have their captive design centres in India. And, of course, we have lots of outsourcing industries. Our design house programme caters to both kinds of categories—people developing products for the Indian market and India developing products for others.

How do you categorise the Indian embedded market?

We cater to more than 30 segments in embedded space. However, in India, there are five primary segments that have specialised focus: point-of-sale (POS) systems for the retail market, digital signage, medical devices, portable handheld devices and automotive devices. These are growing very rapidly and seeing a lot of development activity.

We see a lot of design as well as consumption activity in POS systems for the retail market. So we have a big focus on retail.

When you go to a mall, there are big LCD screens running advertisements. These have a motherboard at the back and a software application running to display advertisements and measure customer footfall. There is lot of innovation in this digital signage area.

If you go to a lab, you will see an X-ray machine, CT scanner, MRI scanner and lots of other diagnostic equipment. We are working on the next generation of these medical devices, including portable X-ray machines, portable ultrasound devices and lots of pathology lab devices.

Coming to portable handheld devices, it's not the cellphones that we are talking about here. These are handheld devices meant for specific applications; for example, if you go to a restaurant (abroad), you will come across a portable device used by the staff to take the order.

In the automotive space, we are looking at embedded

devices from the infotainment side. This is one of the rapidly growing areas.

What are the key technology trends that are shaping the future of chips for embedded systems?

The biggest trend is that embedded devices are beginning to get connected to the Internet. For example, in the past, a retail POS system was a standalone device. However, today POS devices in a store are usually networked and controlled by a central office, so inventory/sales information can be downloaded in real time for taking decisions about advertisements and other promotional schemes.

IDC estimates that 15 billion devices will be connected to the Internet by the year 2015.

What are the top design considerations for embedded developers?

The first thing to consider is performance. Many companies come to us saying "Oh, I wish I had selected 1.8GHz Core2Duo version, instead of 1.2GHz." People don't realise that the performance requirement goes up over time, and have been using the same chip and motherboard for seven years. In the embedded space, the average lifecycle is seven years. However, the software keeps on changing.

The second consideration is reuse. If you are developing a product, you already have a software written for the system. Rewriting it for another architecture is a tedious task. A lot of our customers prefer x86 architecture because a lot of applications are already available on x86. You can reuse them in the embedded space.

Third is availability. Many people don't realise that the single biggest challenge is availability, i.e., when they are ready to place an order for a chip, the chip is unavailable in the market. The design cycles are very long in the embedded space. We offer a lifecycle of minimum seven years. So if the chip is launched today, it will be on our roadmap for at least seven years. Many a times customers start their design three years after the chip is launched. And then they take a year and a half to design the product. And by the time they are ready to order the chip, it's towards the